

Superelevation Calculation Report

Project summary and curve-by-curve transition review

PROJECT	ROUTE	ALIGNMENT	CURVES
Not provided	SR9	SR9	5

Governing standard

MDOT	Mississippi Department of Transportation MDOT superelevation criteria Applicable standards: MDOT STD. DWG SE-2A / MDOT STD. DWG SE-3A
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Curve index

#	Curve	Direction	PC - PT	Speed	Radius	e	Status
1	Curve 1	Right	1456+68.845 - 1482+64.419	65 mph	5,655 ft	0.0340	Calculated
2	Curve 2	Left	1493+78.219 - 1499+59.391	65 mph	3,500 ft	0.0520	Calculated
3	Curve 3	Right	1502+84.391 - 1508+65.563	65 mph	3,500 ft	0.0520	Calculated
4	Curve 4	Right	1526+96.224 - 1532+16.129	65 mph	22,000 ft	0.0000	Normal crown
5	Curve 5	Left	1536+88.988 - 1233+15.920	65 mph	2,500 ft	0.0700	Calculated

Curve 01 / Curve 1

SR9 | SR9 | Right curve

SUPERELEVATION RATE	RUNOFF LENGTH	TANGENT RUNOUT	TOTAL TRANSITION
0.0340 ft/ft	95.00 ft	56.00 ft	151.00 ft

Inputs

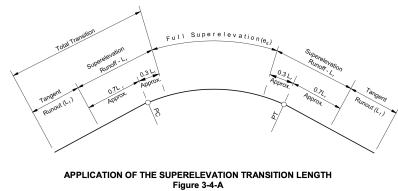
PC station	1456+68.845
PT station	1482+64.419
Design speed	65 mph
Curve radius	5,654.58 ft
Facility / area	centerline / rural
Lane configuration	12.00 ft / 2 rotated
Rate source	Superelevation (table)

Station references

Point of normal crown	1455+46.345
Point of reverse crown	1456+02.345
Full super near PC	1456+97.345
Full super near PT	1482+35.919
Reverse crown out	1483+30.919
Normal crown out	1483+86.919

Criteria reference

Applicable: MDOT STD. DWG SE-2A / MDOT STD. DWG SE-3A
Sources: Crown thresholds - MDOT Table 3-4-A; Superelevation rate - MDOT STD. DWG SE-2A; Runoff length - MDOT STD. DWG SE-3A; Tangent runout - MDOT STD. DWG SE-3A



Reference sheet SE-2A



Reference sheet SE-3A



Notes and provenance

Calculation: Using Table 3-4-C by design speed.; Interpolated from Table 3-4-A.
Curve: Not provided
Design standard: MDOT superelevation criteria

Lane transitions

LEFT LANE			
POINT	STATION	SLOPE	NOTE
NC	1455+46.345	-2.00	Sta = PC - 0.7L - Lt
0%	1456+02.345	0.00	Sta = PC - 0.7L
2%	1456+58.345	+2.00	Sta = PC - 0.7L + Lt
PC	1456+68.845	+2.38	70% super
FULL SUPER	1456+97.345	+3.40	Sta = PC + 0.3L
FULL SUPER	1482+35.919	+3.40	Sta = PT - 0.3L
PT	1482+64.419	+2.38	70% super
0%	1483+30.919	0.00	Sta = PT + 0.7L
NC	1483+86.919	-2.00	0% + runout

RIGHT LANE			
POINT	STATION	SLOPE	NOTE
NC	1456+58.345	-2.00	Sta = PC - 0.7L + Lt
PC	1456+68.845	-2.38	70% super
FULL SUPER	1456+97.345	-3.40	Sta = PC + 0.3L
FULL SUPER	1482+35.919	-3.40	Sta = PT - 0.3L
PT	1482+64.419	-2.38	70% super
NC	1482+74.919	-2.00	Sta = PT + 0.7L - Lt

Curve 02 / Curve 2

SR9 | SR9 | Left curve

SUPERELEVATION RATE	RUNOFF LENGTH	TANGENT RUNOUT	TOTAL TRANSITION
0.0520 ft/ft	145.00 ft	56.00 ft	201.00 ft

Inputs

PC station	1493+78.219
PT station	1499+59.391
Design speed	65 mph
Curve radius	3,500.00 ft
Facility / area	centerline / rural
Lane configuration	12.00 ft / 2 rotated
Rate source	Superelevation (table)

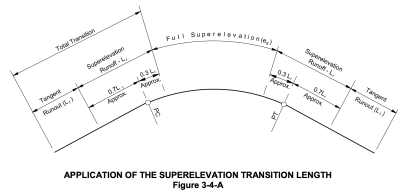
Station references

Point of normal crown	1492+20.719
Point of reverse crown	1492+76.719
Full super near PC	1494+21.719
Full super near PT	1499+15.891
Reverse crown out	1500+60.891
Normal crown out	1501+16.891

Criteria reference

Applicable: MDOT STD. DWG SE-2A / MDOT STD. DWG SE-3A

Sources: Crown thresholds - MDOT Table 3-4-A; Superelevation rate - MDOT STD. DWG SE-2A; Runoff length - MDOT STD. DWG SE-3A; Tangent runout - MDOT STD. DWG SE-3A



Reference sheet SE-2A



Reference sheet SE-3A



Notes and provenance

Calculation: Using Table 3-4-C by design speed.; Interpolated from Table 3-4-A.; Using SE-3A e=0.052 row for e=0.052.

Curve: Not provided

Design standard: MDOT superelevation criteria

Lane transitions

LEFT LANE			
POINT	STATION	SLOPE	NOTE
NC	1493+32.719	-2.00	Sta = PC - 0.7L + Lt
PC	1493+78.219	-3.64	70% super
FULL SUPER	1494+21.719	-5.20	Sta = PC + 0.3L
FULL SUPER	1499+15.891	-5.20	Sta = PT - 0.3L
PT	1499+59.391	-3.64	70% super
NC	1500+04.891	-2.00	Sta = PT + 0.7L - Lt

RIGHT LANE			
POINT	STATION	SLOPE	NOTE
NC	1492+20.719	-2.00	Sta = PC - 0.7L - Lt
0%	1492+76.719	0.00	Sta = PC - 0.7L
2%	1493+32.719	+2.00	Sta = PC - 0.7L + Lt
PC	1493+78.219	+3.64	70% super
FULL SUPER	1494+21.719	+5.20	Sta = PC + 0.3L
FULL SUPER	1499+15.891	+5.20	Sta = PT - 0.3L
PT	1499+59.391	+3.64	70% super
0%	1500+60.891	0.00	Sta = PT + 0.7L
NC	1501+16.891	-2.00	0% + runout

Curve 03 / Curve 3

SR9 | SR9 | Right curve

SUPERELEVATION RATE	RUNOFF LENGTH	TANGENT RUNOUT	TOTAL TRANSITION
0.0520 ft/ft	145.00 ft	56.00 ft	201.00 ft

Inputs

PC station	1502+84.391
PT station	1508+65.563
Design speed	65 mph
Curve radius	3,500.00 ft
Facility / area	centerline / rural
Lane configuration	12.00 ft / 2 rotated
Rate source	Superelevation (table)

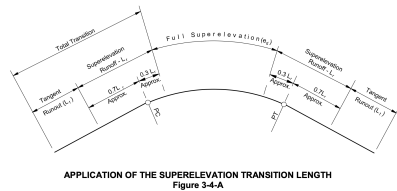
Station references

Point of normal crown	1501+26.891
Point of reverse crown	1501+82.891
Full super near PC	1503+27.891
Full super near PT	1508+22.063
Reverse crown out	1509+67.063
Normal crown out	1510+23.063

Criteria reference

Applicable: MDOT STD. DWG SE-2A / MDOT STD. DWG SE-3A

Sources: Crown thresholds - MDOT Table 3-4-A; Superelevation rate - MDOT STD. DWG SE-2A; Runoff length - MDOT STD. DWG SE-3A; Tangent runout - MDOT STD. DWG SE-3A



Reference sheet SE-2A



Reference sheet SE-3A



Notes and provenance

Calculation: Using Table 3-4-C by design speed.; Interpolated from Table 3-4-A.; Using SE-3A e=0.052 row for e=0.052.

Curve: Not provided

Design standard: MDOT superelevation criteria

Lane transitions

LEFT LANE			
POINT	STATION	SLOPE	NOTE
NC	1501+26.891	-2.00	Sta = PC - 0.7L - Lt
0%	1501+82.891	0.00	Sta = PC - 0.7L
2%	1502+38.891	+2.00	Sta = PC - 0.7L + Lt
PC	1502+84.391	+3.64	70% super
FULL SUPER	1503+27.891	+5.20	Sta = PC + 0.3L
FULL SUPER	1508+22.063	+5.20	Sta = PT - 0.3L
PT	1508+65.563	+3.64	70% super
0%	1509+67.063	0.00	Sta = PT + 0.7L
NC	1510+23.063	-2.00	0% + runout

RIGHT LANE			
POINT	STATION	SLOPE	NOTE
NC	1502+38.891	-2.00	Sta = PC - 0.7L + Lt
PC	1502+84.391	-3.64	70% super
FULL SUPER	1503+27.891	-5.20	Sta = PC + 0.3L
FULL SUPER	1508+22.063	-5.20	Sta = PT - 0.3L
PT	1508+65.563	-3.64	70% super
NC	1509+11.063	-2.00	Sta = PT + 0.7L - Lt

Curve 04 / Curve 4

SR9 | SR9 | Right curve

SUPERELEVATION RATE	RUNOFF LENGTH	TANGENT RUNOUT	TOTAL TRANSITION
0.0000 ft/ft	0.00 ft	0.00 ft	0.00 ft

REVIEW NOTICE

- Normal crown is maintained; no superelevation transition is required.

Inputs

PC station	1526+96.224
PT station	1532+16.129
Design speed	65 mph
Curve radius	22,000.00 ft
Facility / area	centerline / rural
Lane configuration	12.00 ft / 2 rotated
Rate source	Normal crown

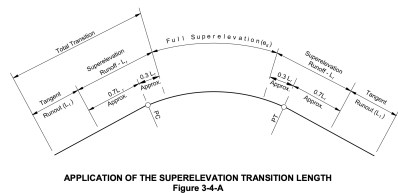
Station references

Normal crown begins	1526+96.224
Normal crown ends	1532+16.129

Criteria reference

Applicable: MDOT STD. DWG SE-2A / MDOT STD. DWG SE-3A

Sources: Crown thresholds - MDOT Table 3-4-A; Superelevation rate - MDOT STD. DWG SE-2A; Transition - Normal crown - no transition required



Reference sheet SE-2A



Reference sheet SE-3A



Notes and provenance

Calculation: Using Table 3-4-C by design speed.; Interpolated from Table 3-4-A.; Interpolated by speed.

Curve: Not provided

Design standard: MDOT superelevation criteria

Lane transitions

LEFT LANE			
POINT	STATION	SLOPE	NOTE
NC	1526+96.224	-2.00	Normal crown maintained; no superelevation transition required
NC	1532+16.129	-2.00	Normal crown maintained through curve

RIGHT LANE			
POINT	STATION	SLOPE	NOTE
NC	1526+96.224	-2.00	Normal crown maintained; no superelevation transition required
NC	1532+16.129	-2.00	Normal crown maintained through curve

Curve 05 / Curve 5

SR9 | SR9 | Left curve

SUPERELEVATION RATE	RUNOFF LENGTH	TANGENT RUNOUT	TOTAL TRANSITION
0.0700 ft/ft	195.00 ft	56.00 ft	251.00 ft

Inputs

PC station	1536+88.988
PT station	1233+15.920
Design speed	65 mph
Curve radius	2,500.00 ft
Facility / area	centerline / rural
Lane configuration	12.00 ft / 2 rotated
Rate source	Superelevation (table)

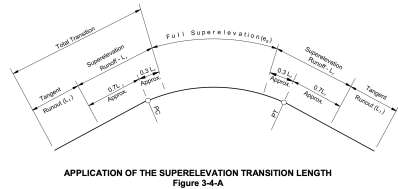
Station references

Point of normal crown	1534+96.488
Point of reverse crown	1535+52.488
Full super near PC	1537+47.488
Full super near PT	1542+93.903
Reverse crown out	1234+52.420
Normal crown out	1235+08.420

Criteria reference

Applicable: MDOT STD. DWG SE-2A / MDOT STD. DWG SE-3A

Sources: Crown thresholds - MDOT Table 3-4-A; Superelevation rate - MDOT STD. DWG SE-2A; Runoff length - MDOT STD. DWG SE-3A; Tangent runout - MDOT STD. DWG SE-3A



Reference sheet SE-2A



Reference sheet SE-3A



Notes and provenance

Calculation: Using Table 3-4-C by design speed.; Interpolated from Table 3-4-A.

Curve: Not provided

Design standard: MDOT superelevation criteria

Lane transitions

LEFT LANE			
POINT	STATION	SLOPE	NOTE
NC	1536+08.488	-2.00	Sta = PC - 0.7L + Lt
PC	1536+88.988	-4.90	70% super
FULL SUPER	1537+47.488	-7.00	Sta = PC + 0.3L
FULL SUPER	1542+93.903	-7.00	Sta = PT - 0.3L
PT	1233+15.920	-4.90	70% super
NC	1233+96.420	-2.00	Sta = PT + 0.7L - Lt

RIGHT LANE			
POINT	STATION	SLOPE	NOTE
NC	1534+96.488	-2.00	Sta = PC - 0.7L - Lt
0%	1535+52.488	0.00	Sta = PC - 0.7L
2%	1536+08.488	+2.00	Sta = PC - 0.7L + Lt
PC	1536+88.988	+4.90	70% super
FULL SUPER	1537+47.488	+7.00	Sta = PC + 0.3L
FULL SUPER	1542+93.903	+7.00	Sta = PT - 0.3L
PT	1233+15.920	+4.90	70% super
0%	1234+52.420	0.00	Sta = PT + 0.7L
NC	1235+08.420	-2.00	0% + runoff